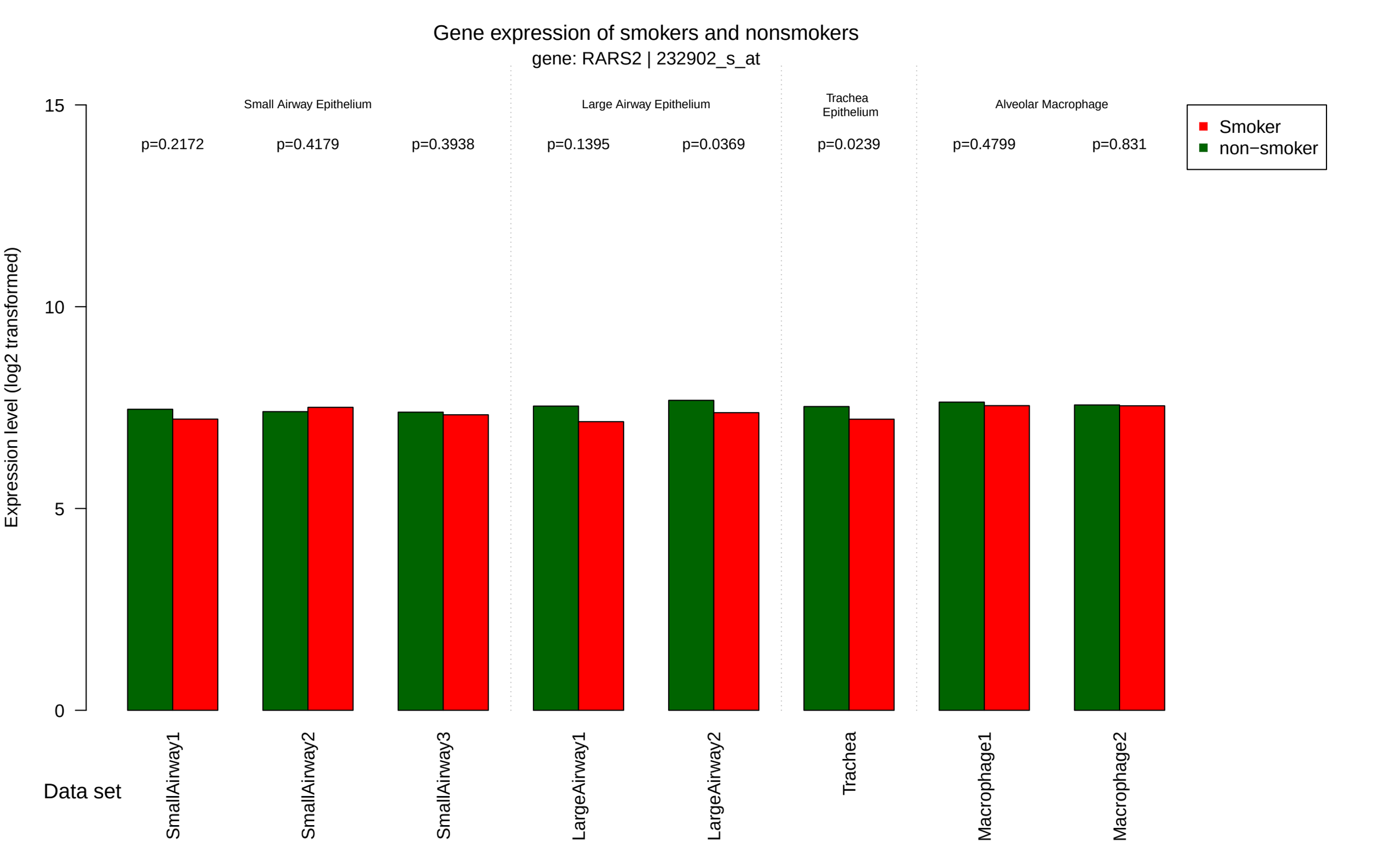
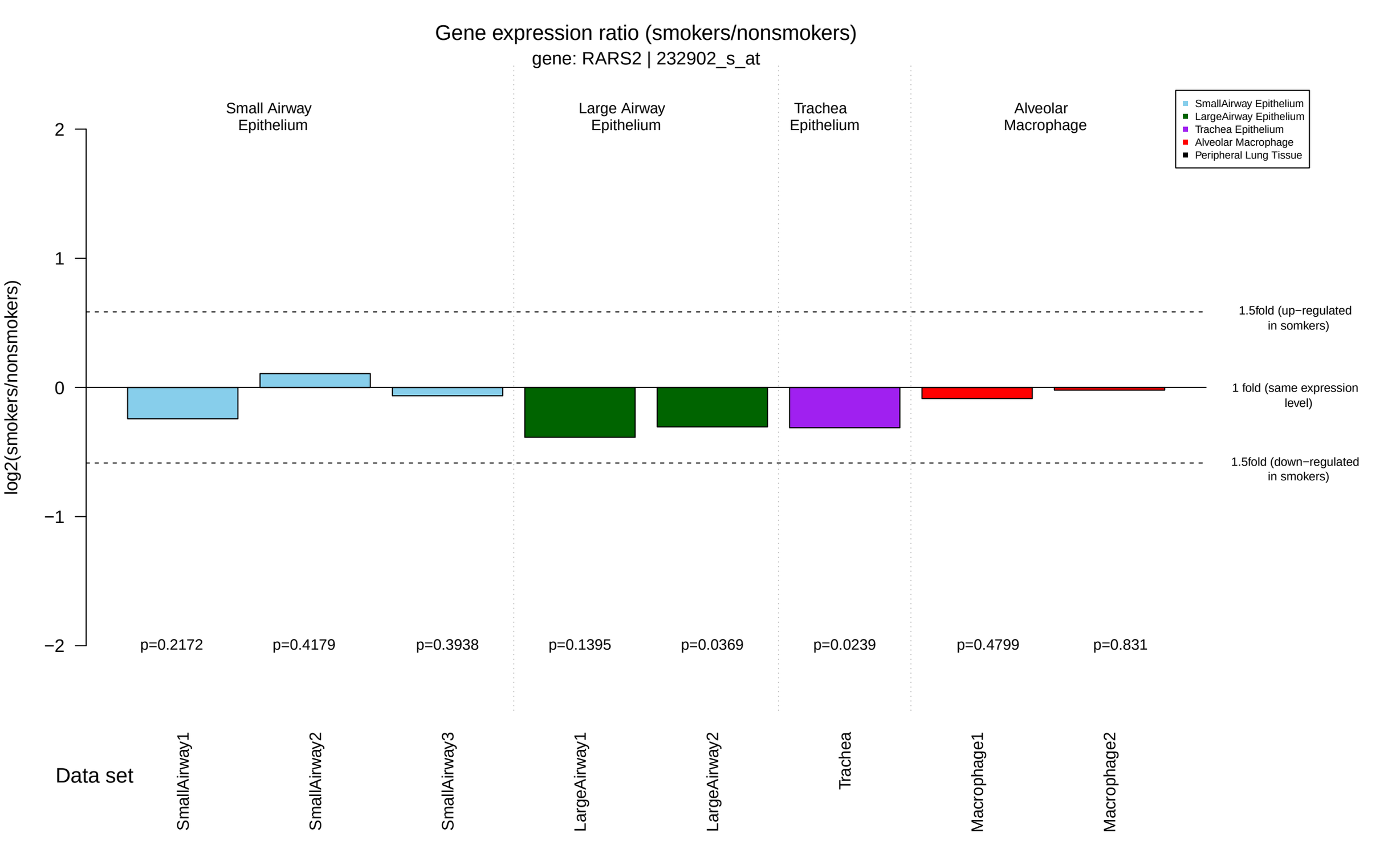
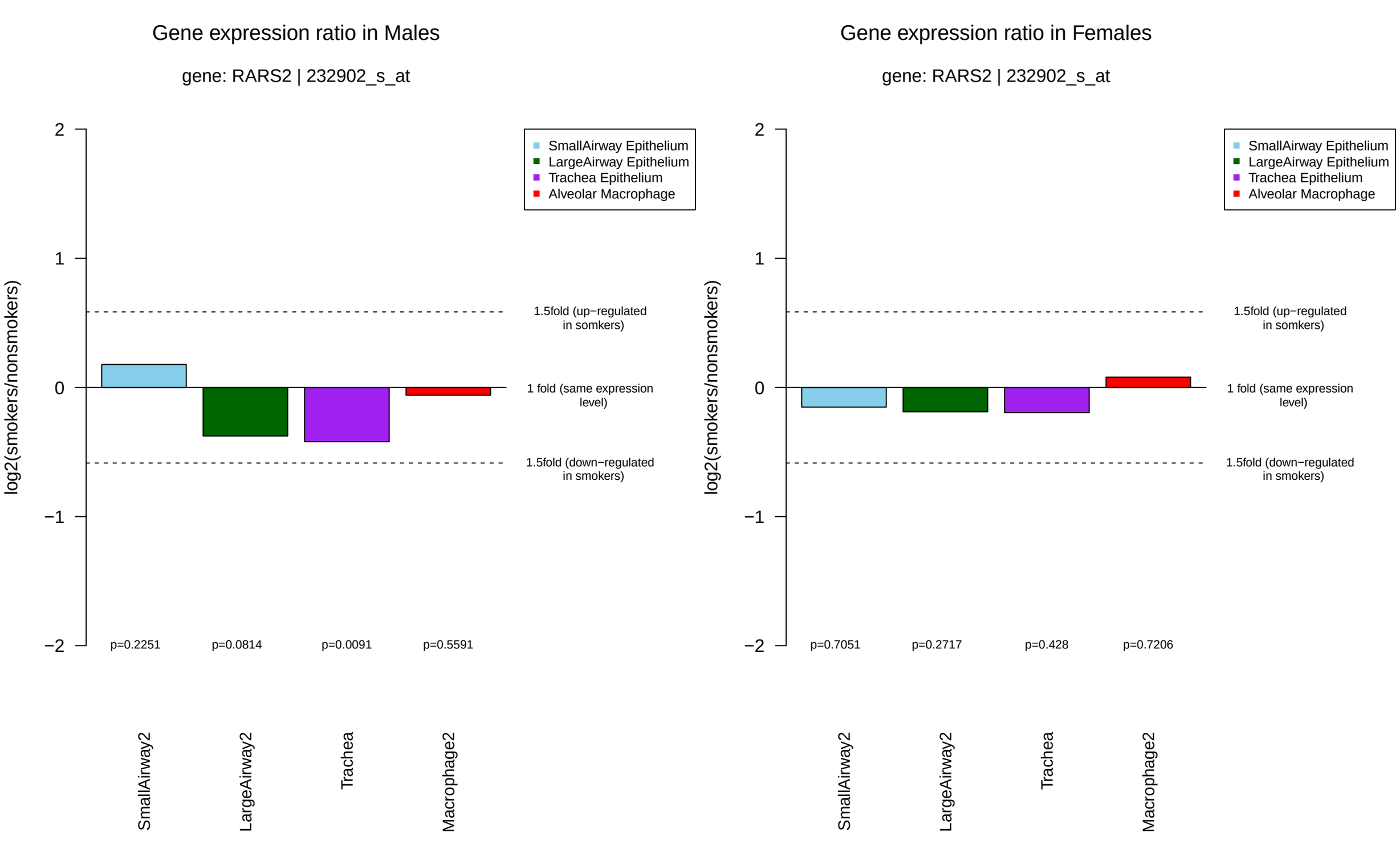
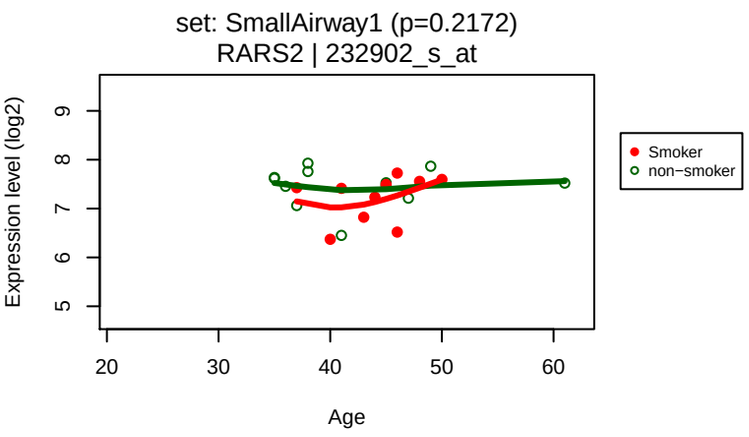
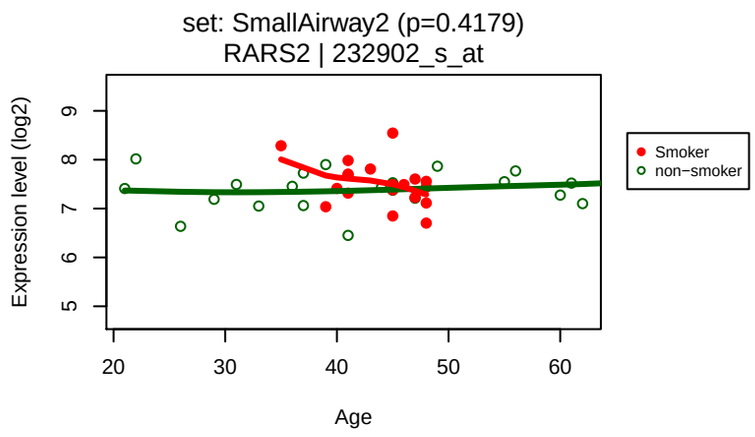
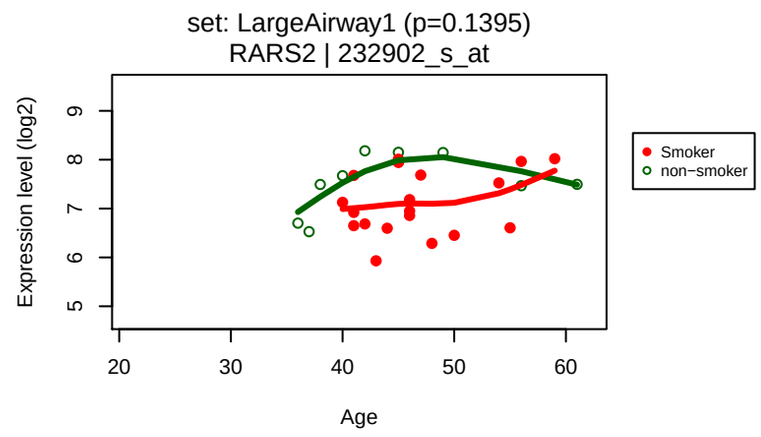

Gene expression of smokers and nonsmokers
gene: RARS2 | 232902_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2172
p=0.4179
p=0.3938
p=0.1395
p=0.0369
p=0.0239
p=0.4799
p=0.831
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

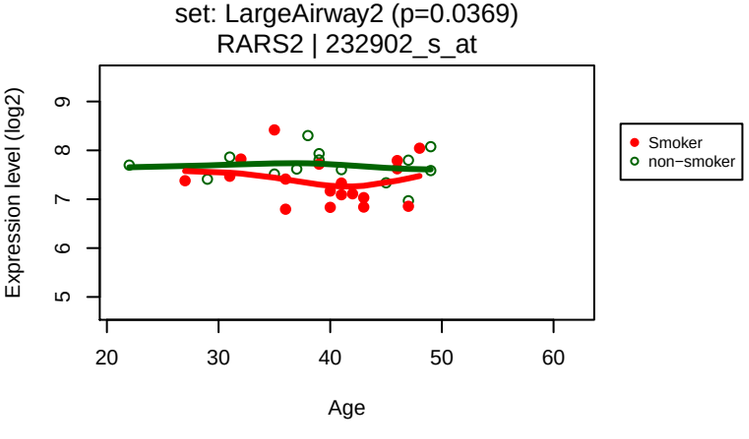

Gene expression ratio (smokers/nonsmokers)
gene: RARS2 | 232902_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2172
p=0.4179
p=0.3938
p=0.1395
p=0.0369
p=0.0239
p=0.4799
p=0.831
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

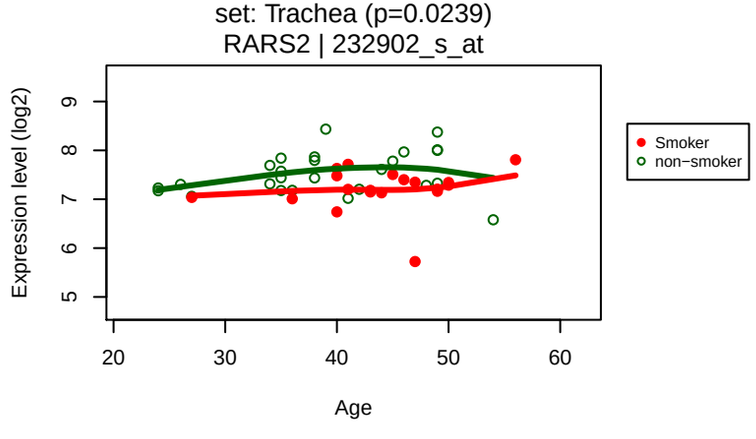

Gene expression ratio in Males
gene: RARS2 | 232902_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2251
p=0.0814
p=0.0091
p=0.5591
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARS2 | 232902_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7051
p=0.2717
p=0.428
p=0.7206
SmallAirway2
LargeAirway2
Trachea
Macrophage2

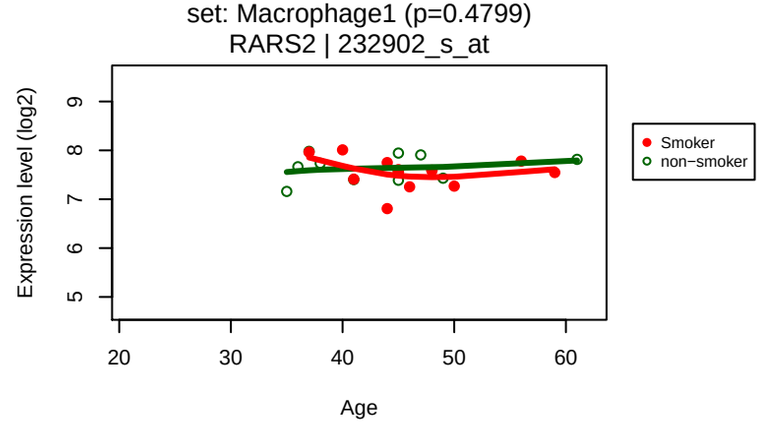

set: SmallAirway1 (p=0.2172)
RARS2 | 232902_s_at
Expression level (log2)
Age
Smoker
non−smoker

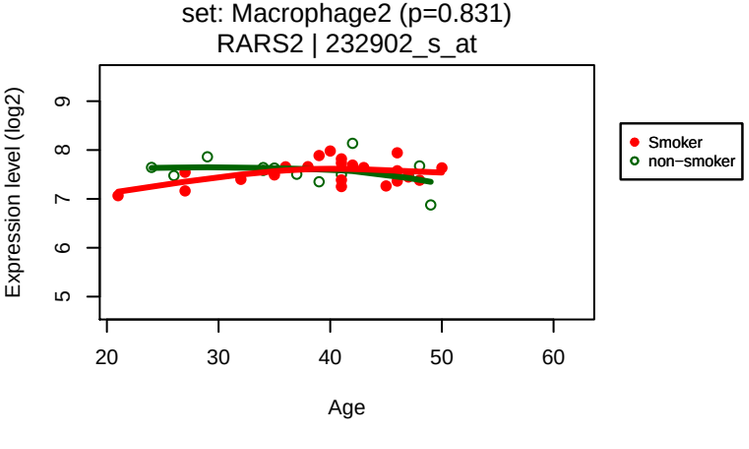

set: SmallAirway2 (p=0.4179)
RARS2 | 232902_s_at
Expression level (log2)
Age
Smoker
non−smoker

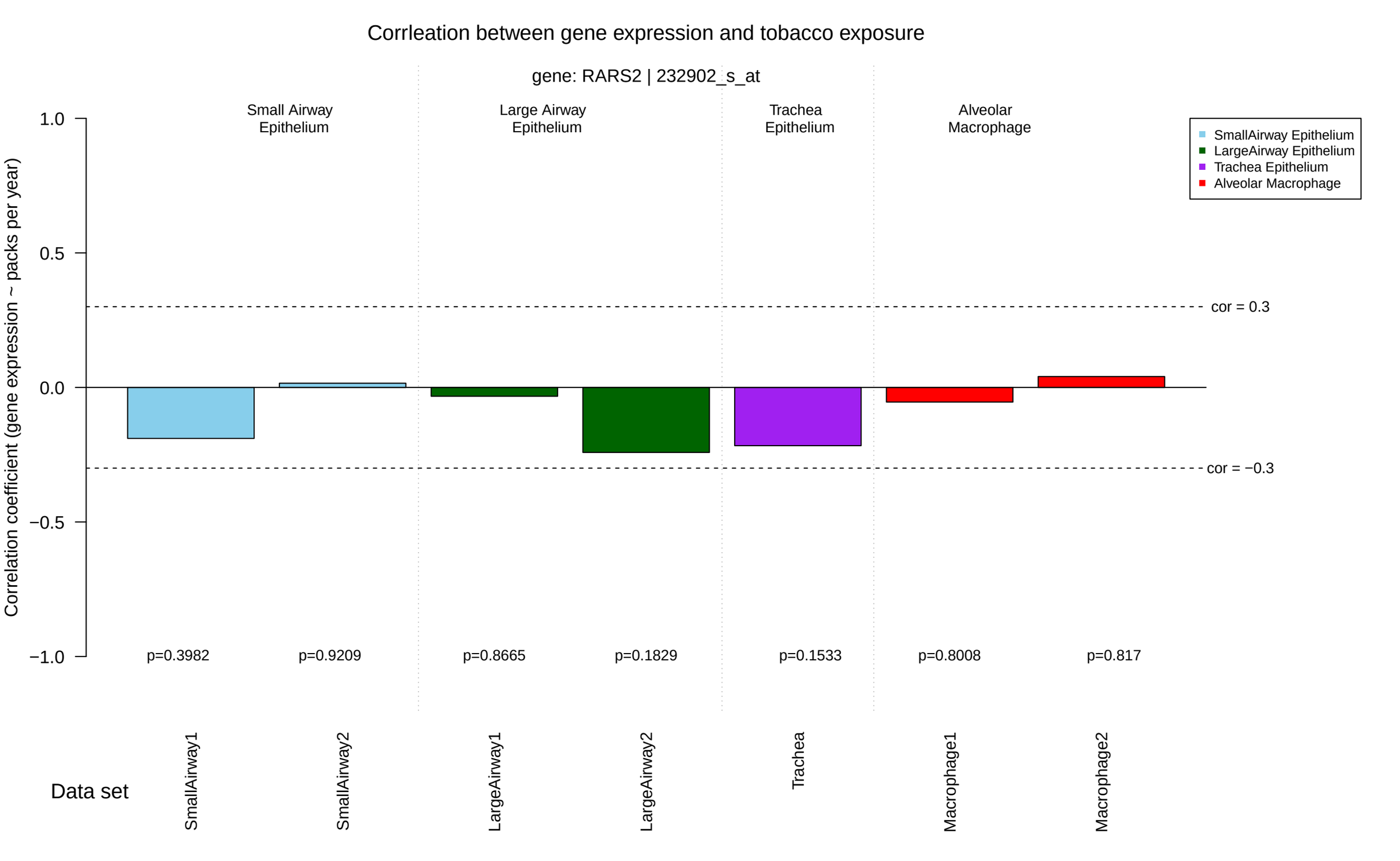

set: LargeAirway1 (p=0.1395)
RARS2 | 232902_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0369)
RARS2 | 232902_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0239)
RARS2 | 232902_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4799)
RARS2 | 232902_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.831)
RARS2 | 232902_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARS2 | 232902_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3982
p=0.9209
p=0.8665
p=0.1829
p=0.1533
p=0.8008
p=0.817
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2